Software Testing Strategies & Quality Assurance

RCS 305 ? Software Engineering | Riara University Revision Resource

1. QA vs. Testing

Quality Assurance (QA) is process-oriented, focusing on defect prevention across all engineering workflows. Testing is product-oriented, focusing on defect identification through dynamic execution.

2. The Modern Test Pyramid

? Unit Tests (Base): High volume, fast execution, isolated tests exercising individual functions or classes with test doubles (mocks/stubs).
? Integration Tests (Middle): Verifies database transactions, API contract boundaries, and inter-service messaging.
? System / End-to-End Tests (Peak): Simulates authentic user workflows through browser UI or API gateway under realistic network conditions.

3. Black-Box vs. White-Box Methodologies

? Black-Box Testing: Evaluates external functionality without visibility into internal source code (Equivalence Partitioning, Boundary Value Analysis, State Transition testing).
? White-Box Testing: Employs code structure knowledge to achieve Statement, Branch, and Condition Coverage benchmarks.

MaxLabs Academic Revision Document -- Copyright Riara University Design Thinking Prototype 2026